

Week 3 Overview and Learning Objectives

Data Structures and Data Sources

In this stage of the course, we pivot from the conceptual value of information to the technical architecture that supports it: **Data Structures and Data Sources**. For leaders in business, public policy, and healthcare, the ability to distinguish between **structured data**—the organized rows of a financial ledger or billing system—and **unstructured data**, such as clinician notes or social media sentiment, is fundamental to defining the scope of any analytical project. Understanding these differences allows managers to move beyond viewing data as a monolithic resource and instead recognize it as a diverse asset that requires specific handling. By mastering **data representation formats** like CSV, JSON, and XML, you will be equipped to select the right container for your data, ensuring it remains accessible and accurate across different organizational platforms.

Effective leadership also requires a deep understanding of where this information originates and how it flows into your decision-making pipeline. We will explore and articulate the primary conduits of organizational intelligence, ranging from the tabular familiarity of **spreadsheets** and the relational integrity of **databases** to the real-time connectivity of **APIs**. This week, we bridge the gap between technical organization and strategic utility, illustrating how the way data is structured directly dictates the quality, speed, and reliability of **managerial decision-making**. Whether you are optimizing a supply chain, evaluating a public health initiative, or managing hospital operations, your ability to align data structures with organizational goals will be the critical factor in turning raw information into competitive advantage.

Key Takeaways for Your Professional Track:

- **MBA:** Leverage APIs to integrate real-time market trends with internal sales databases.
- **MPA:** Convert unstructured citizen feedback into structured themes for policy evaluation.
- **MSHCA:** Transition from unstructured EHR narratives to structured data for population health metrics.

Learning Objectives

Once you complete this your study of this topic, you should be able to :

- Explain differences between structured vs. unstructured data
- Explore and articulate primary sources of data (databases, spreadsheets, APIs)
- Apply appropriate data representation formats

•Connect data structures to decision-making

To help you determine which tool to use for a given organizational challenge, the table below compares **Databases**, **Spreadsheets**, and **APIs** based on their technical strengths and their utility in high-stakes management environments.

Comparative Matrix: Selecting the Right Data Source

Feature	Databases (SQL/NoSQL)	Spreadsheets (Excel/Sheets)	APIs (Application Programming Interfaces)
Best For	Storing vast amounts of historical, relational records.	Quick, "one-off" analysis and simple financial modeling.	Real-time data retrieval and connecting different software.
Data Integrity	High: Uses "Constraints" to prevent duplicate or invalid data entry.	Low: Easy to accidentally overwrite formulas or enter "text" in a "number" cell.	High: Data is delivered in a standardized, programmatic format.
Volume	Millions to billions of rows.	Limited (e.g., ~1M rows in Excel).	Varies; designed for constant, small "streams" of data.
Collaboration	Multi-user; many people can read/write simultaneously.	Limited; version control issues are common ("Final_v2_edit.xlsx").	High; allows different apps (like Salesforce and Epic) to talk to each other.
Common Format	Relational Tables	.xlsx, .csv	JSON, XML